Md Touhidul Islam

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Personal Website LinkedIn GitHub Google Scholar Portfolio

SUMMARY

8+ years building Distributed, ML, and Full-stack systems (**3+** in industry, **5+** in PhD). PhD research: Accessibility, Computer Vision, Applied ML, HCI. Penn State Teaching Fellow, **Distributed Systems**. First-author, **award-winning** papers at top-tier venues.

EDUCATION

Ph.D. in Informatics	College of IST, Penn State University	GPA: 3.97/4	**Jan 2021 – Aug 2026 (Expected)**
M.Sc. in Informatics	College of IST, Penn State University	GPA: 3.97/4	Jan 2021 – Feb 2024
B.Sc. in Computer Science & Engg	Bangladesh University of Engineering & Technology	GPA: 3.91/4	Feb 2013 – Sept 2017

SELECTED WORK EXPERIENCE

Software Engineer **Nov 2017 - Aug 2019**
Reve Systems *Dhaka, Bangladesh*

– Built core modules for a high-throughput distributed telecom billing system, enabling **20k+** daily transactions with **40%** lower latency than the legacy system—through schema/index optimization and Kafka-based asynchronous processing

– Developed a Bengali screen reader with custom OCR + TTS pipeline, improving word-level accuracy by **16%** vs. baseline

Research and Development Engineer **Sept 2019 – Dec 2020**
Bangladesh University of Engineering and Technology *Dhaka, Bangladesh*

– Built a distributed blockchain storage system with containerized microservices, reducing per-node storage by **50%** (**ICBC'20***)

– Built a blockchain-based inventory system for ministries, enabling transparent, auditable tracking of inventory changes

Software Engineering Intern **May 2023 – August 2023**
The Center for Immersive Experiences, Penn State *University Park, PA, USA*

– Built **OpenAccAPI**, an open-source Unity plugin exposing UI and scene elements to assistive technologies — supporting TTS, UI highlighting, non-visual navigation, cursor feedback; retrofitted injection into existing apps without source access [GitHub]

– Validated across **15** popular Unity/Steam titles; designed an injector that inserts the plugin into existing games within **3** clicks

Software Engineering Intern **May 2024 – August 2024**
The Center for Immersive Experiences, Penn State *University Park, PA, USA*

– Collected 3D/VR game scene data from engine telemetry (no pixels); designed **O**bject **T**rajectory **N**arratives (**OTN**), a structured natural language representation of per-object scene dynamics optimized for LLM consumption

– OTN enables **7B** LLMs to match **GPT-4V-level accuracy** on complex motion and scene narration at a fraction of the compute

Graduate Research Assistant **Jan 2021 – Present**
The Pennsylvania State University *University Park, PA, USA*

– Built and evaluated accessibility-oriented systems for blind and low-vision (BLV) interaction; developed real-world computer vision/LMM reliability methods including evaluation, segmentation, targeted unlearning (details in **Research Projects** section)

TECHNICAL SKILLS

Programming and Scripting: Python, Java, C/C++, C#, Git, MATLAB, JavaScript, Node.js, Express.js, ReactJS

Systems & Cloud: Linux, Docker, Azure, Apache, Redis, Kafka, REST APIs, MySQL, PostgreSQL, AWS

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, LLM fine-tuning, PEFT, vLLM, LangChain, FAISS, MCP

SELECTED RESEARCH PROJECTS

Accessibility, Systems, and Human-Computer Interaction:

– **Wheeler**: Designed a 3-wheel input device for blind and low-vision (BLV) users with two modes – *UI/menu navigation* and *2D mouse-like navigation* – achieving a **40%** reduction in app hierarchy traversal time (Best Paper HM – **UIST'24***)

– **BLV-Nav-Objects**: Identified **90 key objects** essential for BLV individuals' navigation from 21 videos; refined them through a focus group study, revealing major gaps in prominent vision datasets like *KITTI* and *MS-COCO* [GitHub] (**ASSETS'24***)

– **SpaceXMag**: Developed a screen-space optimization framework that improves magnifier efficiency for low-vision users, reducing overview and target-finding task times by **28–43%** in user studies [GitHub] (**IMWUT'23***)

– **Accessibility-Metrics**: Identified navigational complexity as the key usability barrier for blind users; proposed a probabilistic model with **three** metrics to quantify and improve desktop application accessibility [GitHub] (**CHI'23***)

Machine Learning and Computer Vision:

– **Temporal VOS Loss**: Developed a new loss for video object segmentation that enforces smooth object motion, yielding up to **4%** J&F gains on small, occluded, and reappearing objects across *DAVIS*, *YT-VOS*, and *LVOS* (in **ECCV'26** review)

- **IKIWISI**: Developed an interactive vision-language model evaluation tool using heatmap visualizations to reveal inconsistencies and object-tracking failures—supporting scalable debugging [GitHub] (Best Paper HM – **DIS'25***)
- **ForenSight**: A local-first video investigation pipeline using **CLIP embeddings + FAISS** for text-to-timestamp retrieval and event-centric playback; supports ROI/line rules, policy tagging, multi-camera correlation, and case export [GitHub]

Large Language Models and Agentic AI:

- **TeachPilot**: A classroom LLM assistant (Streamlit + Ollama) with instructor-controlled prompts and help levels to promote think-first engagement; **statistically significant** gains in student AI literacy via pre/post study [GitHub] (**AIED'26***)
- **LLM Unlearning with Synthetic Data**: Built LoRA-based PEFT unlearning to remove behaviors (e.g., bias) via multi-GPU fine-tuning with synthetic data–generated via Meta's synthetic-data-kit using vLLM (Llama-3.3-70B) [GitHub]

Papers marked with * are first-author publications. See all my publications here